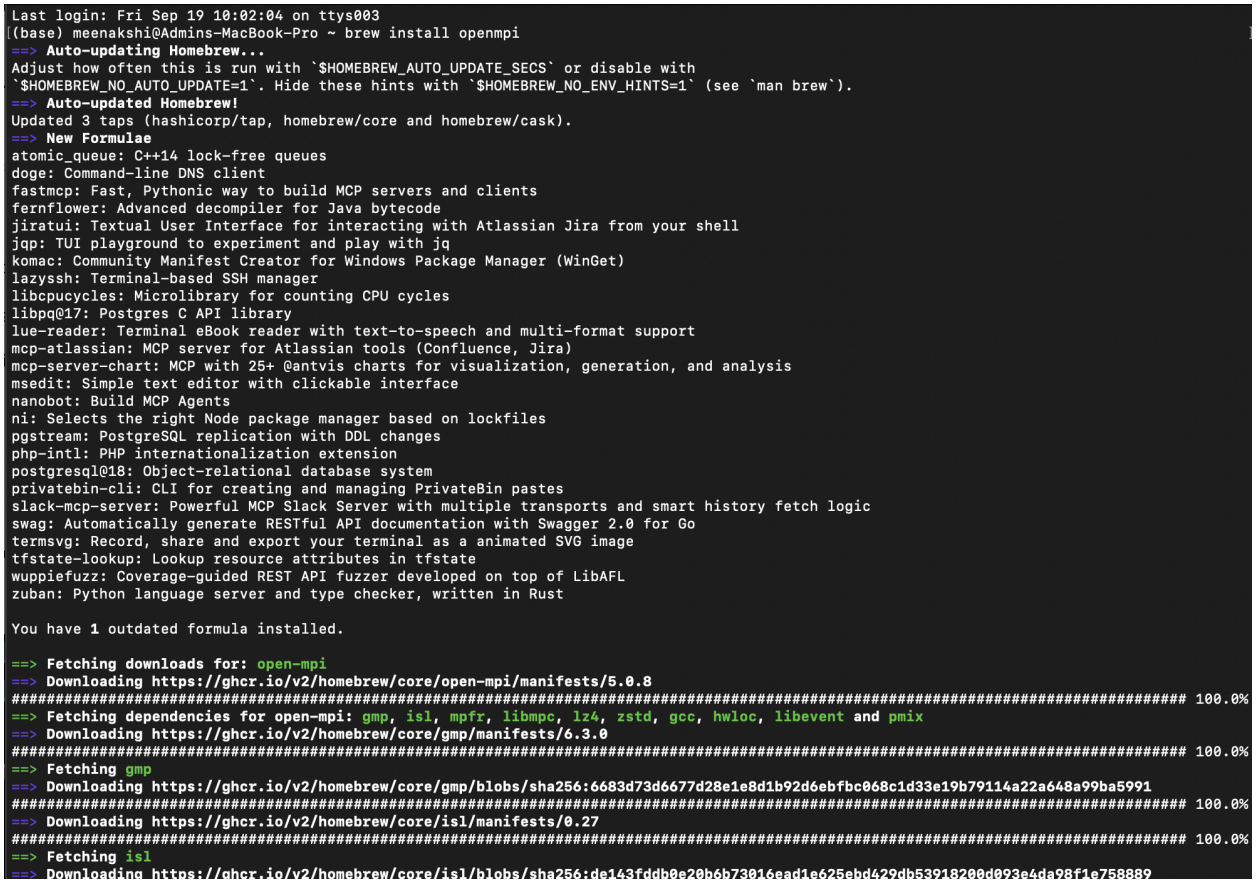
Student ID - 24235415

Cloud Architecture Week 2 lab

Step 1-: Install OpenMPI using Homebrew

brew install openmpi

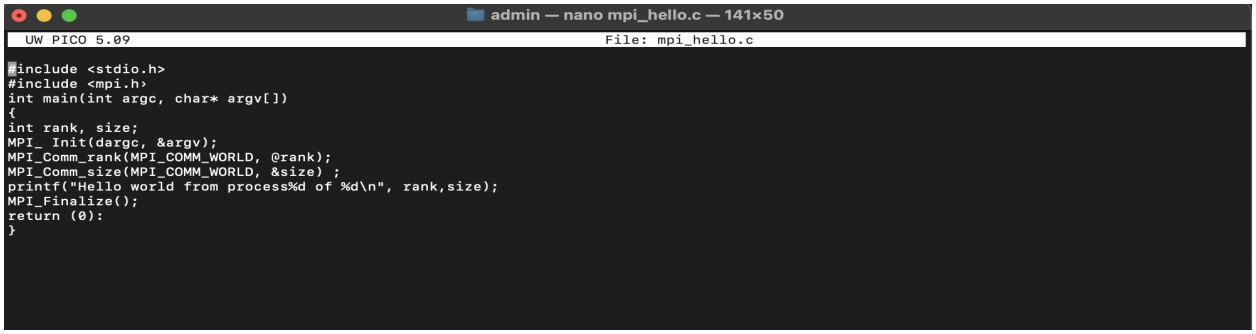


```
==> Downloading https://ghcr.io/v2/homebrew/core/hwloc/manifests/2.12.2
######################################################################################################################################################## 100.0%
==> Fetching hwloc
==> Downloading https://ghcr.io/v2/homebrew/core/hwloc/blobs/sha256:356391e580cf546a4b1b4180fa630a64ba6bd7bf00164914b58e41f0f2fa2cea
######################################################################################################################################################## 100.0%
==> Downloading https://ghcr.io/v2/homebrew/core/libevent/manifests/2.1.12_1
######################################################################################################################################################## 100.0%
==> Fetching libevent
==> Downloading https://ghcr.io/v2/homebrew/core/libevent/blobs/sha256:65fc7c61fec0f5ae0c5dfc8fc7e3b6b0507d3f1c7c308a332802541f00334963
######################################################################################################################################################## 100.0%
==> Downloading https://ghcr.io/v2/homebrew/core/pmix/manifests/5.0.9
######################################################################################################################################################## 100.0%
==> Fetching pmix
==> Downloading https://ghcr.io/v2/homebrew/core/pmix/blobs/sha256:1ee7ba6f1b0345697fb712c5ad5bc85135675dc4d6b61bca1b11cbcc401aea4f
######################################################################################################################################################## 100.0%
==> Fetching open-mpi
==> Downloading https://ghcr.io/v2/homebrew/core/open-mpi/blobs/sha256:ffda541a4f3b0f37e853066cdde31ab459834d0ded4f0e03ab4a74a8dafd6485
######################################################################################################################################################## 100.0%
==> Installing dependencies for open-mpi: hwloc, libevent and pmix
==> Installing open-mpi dependency: hwloc
==> Downloading https://ghcr.io/v2/homebrew/core/hwloc/manifests/2.12.2
Already downloaded: /Users/admin/Library/Caches/Homebrew/downloads/2a9d34b3c517f112c71775959f258112d242e3ed51bc68d83a8537791c60ff95--hwloc-2.
12.2.bottle_manifest.json
==> Pouring hwloc--2.12.2.arm64_sequoia.bottle.tar.gz
🍺  /opt/homebrew/Cellar/hwloc/2.12.2: 976 files, 10.8MB
==> Installing open-mpi dependency: libevent
==> Downloading https://ghcr.io/v2/homebrew/core/libevent/manifests/2.1.12_1
Already downloaded: /Users/admin/Library/Caches/Homebrew/downloads/68b113f9ab63db45f4e1860de522ce2ca4fa081eb3c0d5c7d6005a35c3cf8d06--libevent
-2.1.12_1.bottle_manifest.json
==> Pouring libevent--2.1.12_1.arm64_sequoia.bottle.tar.gz
🍺  /opt/homebrew/Cellar/libevent/2.1.12_1: 58 files, 2.2MB
==> Installing open-mpi dependency: pmix
==> Downloading https://ghcr.io/v2/homebrew/core/pmix/manifests/5.0.9
Already downloaded: /Users/admin/Library/Caches/Homebrew/downloads/960bd2444e6411f613552de8ac01fbd71f326b2870ced60c689a2f36cb4b1603--pmix-5.0
.9.bottle_manifest.json
==> Pouring pmix--5.0.9.arm64_sequoia.bottle.tar.gz
🍺  /opt/homebrew/Cellar/pmix/5.0.9: 385 files, 14.9MB
==> Installing open-mpi
==> Pouring open-mpi--5.0.8.arm64_sequoia.bottle.tar.gz
🍺  /opt/homebrew/Cellar/open-mpi/5.0.8: 2,624 files, 74.7MB
==> Running `brew cleanup open-mpi`...
Disable this behaviour by setting `HOMEBREW_NO_INSTALL_CLEANUP=1`.
Hide these hints with `HOMEBREW_NO_ENV_HINTS=1` (see `man brew`).
==> No outdated dependents to upgrade!
```

Step 2-: Create the MPI Program

```
(base) meenakshi@Admins-MacBook-Pro ~ touch mpi_hello.c
(base) meenakshi@Admins-MacBook-Pro ~ nano mpi_hello.c
```

```
#include <stdio.h>
#include <mpi.h>
int main(int argc, char *argv[])
{
   int rank, size;
   MPI_Init(&argc, &argv);
   MPI_Comm_rank(MPI_COMM_WORLD, &rank);
   MPI_Comm_size(MPI_COMM_WORLD, &size);
   printf("Hello world from process %d of %d\n", rank, size);
   MPI_Finalize();
   return 0;
}
```

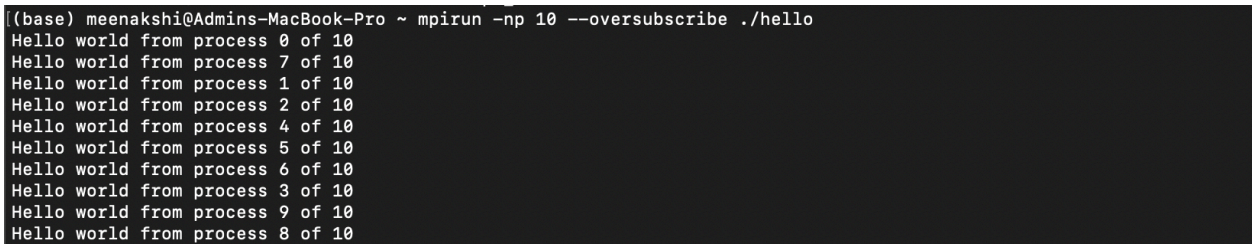


STEP 6: QUESTIONS:

1- **Why are the process numbered out-of-order?**

Ans-: All process gets printed directly to the terminal without any sync as all the operations are running independently , so the operating system decides the order in which the output appears. That is the reason why the line hello world don't always print in sequence.





2-Why do we need to use the oversubscribe parameter?

If we want to try running more processes than the cores available we have to use the oversubscribe parameter. As MPI runs the process independently and expects every process to be completed faster and no process needs to wait for the CPU and make the task slower.

```
Process 1 is working... (iteration 5)
(base) meenakshi@Admins-MacBook-Pro ~ mpirun -np 30 ./hello
--------------------------------------------------------------------------
There are not enough slots available in the system to satisfy the 30
slots that were requested by the application:

  ./hello

Either request fewer procs for your application, or make more slots
available for use.

A "slot" is the PRRTE term for an allocatable unit where we can
launch a process.  The number of slots available are defined by the
environment in which PRRTE processes are run:

  1. Hostfile, via "slots=N" clauses (N defaults to number of
     processor cores if not provided)
  2. The --host command line parameter, via a ":N" suffix on the
     hostname (N defaults to 1 if not provided)
  3. Resource manager (e.g., SLURM, PBS/Torque, LSF, etc.)
  4. If none of a hostfile, the --host command line parameter, or an
     RM is present, PRRTE defaults to the number of processor cores

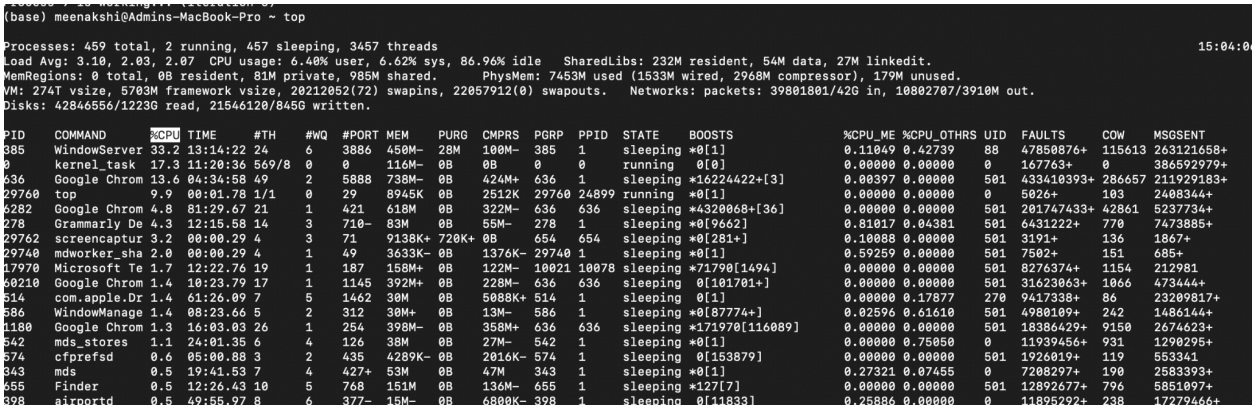
In all the above cases, if you want PRRTE to default to the number
of hardware threads instead of the number of processor cores, use the
--use-hwthread-cpus option.

Alternatively, you can use the --map-by :OVERSUBSCRIBE option to ignore the
number of available slots when deciding the number of processes to
launch.
--------------------------------------------------------------------------
```

```
(base) meenakshi@Admins-MacBook-Pro ~ mpirun -np 30 --oversubscribe ./hello
Hello world from process 5 of 30
Process 5 is working... (iteration 1)
Hello world from process 1 of 30
Process 1 is working... (iteration 1)
Hello world from process 2 of 30
Process 2 is working... (iteration 1)
Hello world from process 3 of 30
Process 3 is working... (iteration 1)
Hello world from process 26 of 30
Process 26 is working... (iteration 1)
Hello world from process 10 of 30
Process 10 is working... (iteration 1)
Hello world from process 28 of 30
Process 28 is working... (iteration 1)
Hello world from process 22 of 30
Process 22 is working... (iteration 1)
Hello world from process 7 of 30
Process 7 is working... (iteration 1)
Hello world from process 13 of 30
Process 13 is working... (iteration 1)
Hello world from process 15 of 30
Process 15 is working... (iteration 1)
Hello world from process 18 of 30
Process 18 is working... (iteration 1)
Hello world from process 9 of 30
Process 9 is working... (iteration 1)
```

3-Can you monitor the utilisation of the system using the top command?

Using the top command in the terminal gives you a real time system monitor.



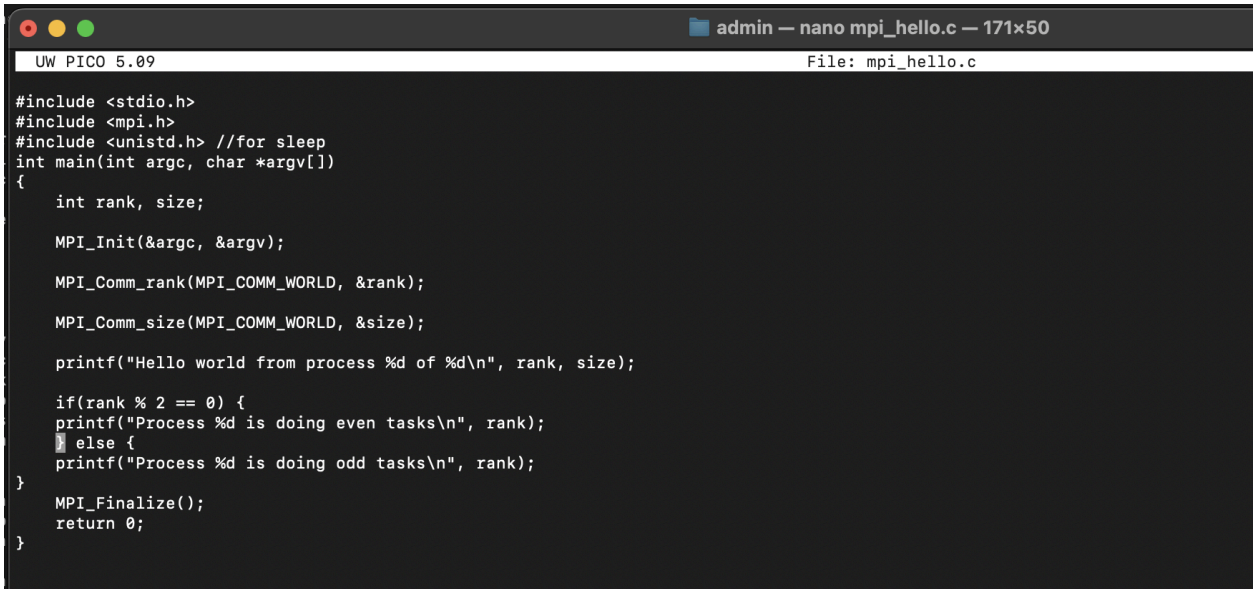
```
(base) meenakshi@Admins-MacBook-Pro ~ top

Processes: 459 total, 2 running, 457 sleeping, 3457 threads                                                                                                15:04:0
Load Avg: 3.10, 2.03, 2.07  CPU usage: 6.40% user, 6.62% sys, 86.96% idle   SharedLibs: 232M resident, 54M data, 27M linkedit.
MemRegions: 0 total, 0B resident, 81M private, 985M shared.      PhysMem: 7453M used (1533M wired, 2968M compressor), 179M unused.
VM: 274T vsize, 5703M framework vsize, 20212052(72) swapins, 22057912(0) swapouts.   Networks: packets: 39801801/42G in, 10802707/3910M out.
Disks: 42846556/1223G read, 21546120/845G written.

PID    COMMAND      %CPU TIME     #TH    #WQ  #PORT MEM    PURG  CMPRS  PGRP  PPID  STATE    BOOSTS            %CPU_ME %CPU_OTHRS UID  FAULTS     COW    MSGSENT
385    WindowServer 33.2 13:14:22 24     6    3886  450M-  28M   100M-  385   1     sleeping *0[1]             0.11049 0.42739    88   47850876+  115613 263121658+
0      kernel_task  17.3 11:20:36 569/8  0    0     116M-  0B    0B     0     0     running   0[0]             0.00000 0.00000    0    167763+    0      386592979+
636    Google Chrom 13.6 04:34:58 49     2    5888  738M-  0B    424M+  636   1     sleeping *16224422+[3]     0.00397 0.00000    501  433410393+ 286657 211929183+
29760  top          9.9  00:01.78 1/1    0    29    8945K  0B    2512K  29760 24899 running  *0[1]             0.00000 0.00000    0    5026+      103    2408344+
6282   Google Chrom 4.8  81:29.67 21     1    421   618M   0B    322M-  636   636   sleeping *4320068+[36]     0.00000 0.00000    501  201747433+ 42861  5237734+
278    Grammarly De 4.3  12:15.58 14     3    710-  83M    0B    55M-   278   1     sleeping *0[9662]          0.81017 0.04381    501  6431222+   770    7473885+
29762  screencaptur 3.2  00:00.29 4      3    71    9138K+ 720K+ 0B     654   654   sleeping *0[281+]          0.10088 0.00000    501  3191+      136    1867+
29740  mdworker_sha 2.0  00:00.29 4      1    49    3633K- 0B    1376K- 29740 1     sleeping *0[1]             0.59259 0.00000    501  7502+      151    685+
17970  Microsoft Te 1.7  12:22.76 19     1    187   158M+  0B    122M-  10021 10078 sleeping *71790[1494]      0.00000 0.00000    501  8276374+   1154   212981
60210  Google Chrom 1.4  10:23.79 17     1    1145  392M+  0B    228M-  636   636   sleeping  0[101701+]       0.00000 0.00000    501  31623063+  1066   473444+
514    com.apple.Dr 1.4  61:26.09 7      5    1462  30M    0B    5088K+ 514   1     sleeping  0[1]             0.00000 0.17877    270  9417338+   86     23209817+
586    WindowManage 1.4  08:23.66 5      2    312   30M+   0B    13M-   586   1     sleeping *0[87774+]        0.02596 0.61610    501  4980109+   242    1486144+
1180   Google Chrom 1.3  16:03.03 26     1    254   398M-  0B    358M+  636   636   sleeping *171970[116089]   0.00000 0.00000    501  18386429+  9150   2674623+
542    mds_stores   1.1  24:01.35 6      4    126   38M    0B    27M-   542   1     sleeping *0[1]             0.00000 0.75050    0    11939456+  931    1290295+
574    cfprefsd     0.6  05:00.88 3      2    435   4289K- 0B    2016K- 574   1     sleeping  0[153879]        0.00000 0.00000    501  1926019+   119    553341
343    mds          0.5  19:41.53 7      4    427+  53M    0B    47M    343   1     sleeping *0[1]             0.27321 0.07455    0    7208297+   190    2583393+
655    Finder       0.5  12:26.43 10     5    768   151M   0B    136M-  655   1     sleeping *127[7]           0.00000 0.00000    501  12892677+  796    5851097+
398    airportd     0.5  49:55.97 8      6    377-  15M-   0B    6800K- 398   1     sleeping  0[11833]         0.25886 0.00000    0    11895292+  238    17279466+
```

Using the top command for MPI Program also gives you the process execution history. Like CPU utilisation ,memory usage etc. However in the executed file there is very less time taken by the CPU so we can't see it directly the history of execution but we can use some other command sleep or waitime in code to check the process utilisation.

4-Can you extend the program to execute longer to see the processor utilisation?



admin — nano mpi_hello.c — 171×50

UW PICO 5.09 File: mpi_hello.c

```
#include <stdio.h>
#include <mpi.h>
#include <unistd.h> //for sleep
int main(int argc, char *argv[])
{
    int rank, size;

    MPI_Init(&argc, &argv);

    MPI_Comm_rank(MPI_COMM_WORLD, &rank);

    MPI_Comm_size(MPI_COMM_WORLD, &size);

    printf("Hello world from process %d of %d\n", rank, size);

    if(rank % 2 == 0) {
    printf("Process %d is doing even tasks\n", rank);
    } else {
    printf("Process %d is doing odd tasks\n", rank);
}
    MPI_Finalize();
    return 0;
}
```

```
(base) meenakshi@Admins-MacBook-Pro ~ mpirun -np 10 --oversubscribe ./hello
Hello world from process 4 of 10
Process 4 is doing even tasks
Hello world from process 1 of 10
Process 1 is doing odd tasks
Hello world from process 3 of 10
Process 3 is doing odd tasks
Hello world from process 2 of 10
Process 2 is doing even tasks
Hello world from process 9 of 10
Process 9 is doing odd tasks
Hello world from process 0 of 10
Process 0 is doing even tasks
Hello world from process 5 of 10
Process 5 is doing odd tasks
Hello world from process 8 of 10
Process 8 is doing even tasks
Hello world from process 6 of 10
Process 6 is doing even tasks
Hello world from process 7 of 10
Process 7 is doing odd tasks
```

```
(base) meenakshi@Admins-MacBook-Pro ~ mpirun -np 10 --oversubscribe ./hello
Hello world from process 1 of 10
Process 1 is working... (iteration 1)
Hello world from process 4 of 10
Process 4 is working... (iteration 1)
Hello world from process 7 of 10
Process 7 is working... (iteration 1)
Hello world from process 2 of 10
Process 2 is working... (iteration 1)
Hello world from process 9 of 10
Process 9 is working... (iteration 1)
Hello world from process 5 of 10
Process 5 is working... (iteration 1)
Hello world from process 8 of 10
Process 8 is working... (iteration 1)
Hello world from process 6 of 10
Process 6 is working... (iteration 1)
Hello world from process 3 of 10
Process 3 is working... (iteration 1)
Hello world from process 0 of 10
Process 0 is working... (iteration 1)
Process 2 is working... (iteration 2)
Process 4 is working... (iteration 2)
Process 1 is working... (iteration 2)
Process 7 is working... (iteration 2)
Process 8 is working... (iteration 2)
Process 3 is working... (iteration 2)
Process 6 is working... (iteration 2)
Process 0 is working... (iteration 2)
Process 5 is working... (iteration 2)
Process 9 is working... (iteration 2)
Process 2 is working... (iteration 3)
Process 0 is working... (iteration 3)
Process 6 is working... (iteration 3)
Process 4 is working... (iteration 3)
Process 7 is working... (iteration 3)
Process 3 is working... (iteration 3)
Process 9 is working... (iteration 3)
Process 8 is working... (iteration 3)
Process 1 is working... (iteration 3)
Process 5 is working... (iteration 3)
Process 6 is working... (iteration 4)
Process 2 is working... (iteration 4)
Process 0 is working... (iteration 4)
Process 4 is working... (iteration 4)
Process 3 is working... (iteration 4)
Process 8 is working... (iteration 4)
Process 5 is working... (iteration 4)
Process 7 is working... (iteration 4)
```